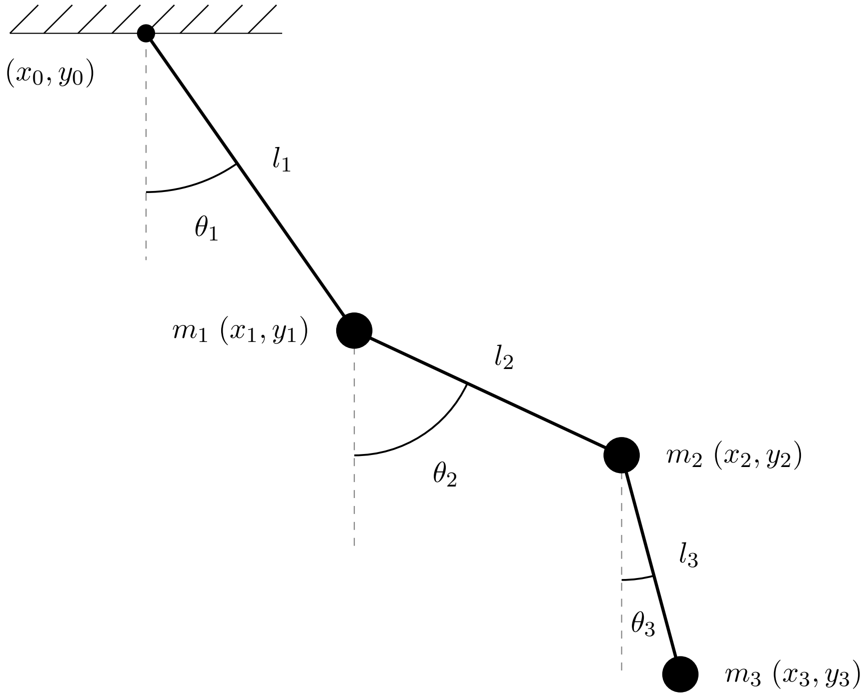
Euler–Lagrange derivation (3-arm pendulum)

Setup

Pivot at the origin. Generalized coordinates: the absolute angles θ_1, θ_2, θ_3. Screen convention: $(0,0)$ is the top-left corner, so y increases downward.



Positions

$$x_1 = x_0 + l_1 \sin\theta_1 \qquad y_1 = y_0 + l_1 \cos\theta_1$$
$$x_2 = x_1 + l_2 \sin\theta_2 \qquad y_2 = y_1 + l_2 \cos\theta_2$$
$$x_3 = x_2 + l_3 \sin\theta_3 \qquad y_3 = y_2 + l_3 \cos\theta_3$$

Velocities

$$\dot{x}_1 = l_1 \cos\theta_1 \dot{\theta}_1 \qquad \dot{y}_1 = -l_1 \sin\theta_1 \dot{\theta}_1$$

$$v_1 = \sqrt{\dot{x}_1^2 + \dot{y}_1^2} \qquad v_1^2 = \dot{x}_1^2 + \dot{y}_1^2$$

$$\dot{x}_2 = \dot{x}_1 + l_2 \cos\theta_2 \dot{\theta}_2 \qquad \dot{y}_2 = \dot{y}_1 - l_2 \sin\theta_2 \dot{\theta}_2$$

$$v_2 = \sqrt{\dot{x}_2^2 + \dot{y}_2^2} \qquad v_2^2 = \dot{x}_2^2 + \dot{y}_2^2$$

$$\dot{x}_3 = \dot{x}_2 + l_3 \cos\theta_3 \dot{\theta}_3 \qquad \dot{y}_3 = \dot{y}_2 - l_3 \sin\theta_3 \dot{\theta}_3$$

$$v_3 = \sqrt{\dot{x}_3^2 + \dot{y}_3^2} \qquad v_3^2 = \dot{x}_3^2 + \dot{y}_3^2$$

Kinetic energy

$$T = \frac{1}{2}mv^2 \qquad \Longrightarrow \qquad T = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2$$

Potential energy

$$V = mgh, \qquad h_1 = -y_1, \quad h_2 = -y_2, \quad h_3 = -y_3$$

$$V = g(m_1h_1 + m_2h_2 + m_3h_3)$$

$$V = -g(m_1y_1 + m_2y_2 + m_3y_3)$$

Lagrangian

$$L = T - V$$

$$L = \frac{1}{2}(m_1v_1^2 + m_2v_2^2 + m_3v_3^2) + g(m_1y_1 + m_2y_2 + m_3y_3)$$

$$L = \frac{1}{2}\left(m_1(\dot{x}_1^2 + \dot{y}_1^2) + m_2(\dot{x}_2^2 + \dot{y}_2^2) + m_3(\dot{x}_3^2 + \dot{y}_3^2)\right) + g(m_1y_1 + m_2y_2 + m_3y_3)$$

Substituting the velocities:

$$\begin{aligned} L = &\frac{1}{2}m_1\left((l_1\cos\theta_1\dot{\theta}_1)^2 + (-l_1\sin\theta_1\dot{\theta}_1)^2\right) \\ &+ \frac{1}{2}m_2\left((\dot{x}_1 + l_2\cos\theta_2\dot{\theta}_2)^2 + (\dot{y}_1 - l_2\sin\theta_2\dot{\theta}_2)^2\right) \\ &+ \frac{1}{2}m_3\left((\dot{x}_2 + l_3\cos\theta_3\dot{\theta}_3)^2 + (\dot{y}_2 - l_3\sin\theta_3\dot{\theta}_3)^2\right) \\ &+ g(m_1y_1 + m_2y_2 + m_3y_3) \end{aligned}$$

$$\begin{aligned} L = &\frac{1}{2}m_1\left((l_1\cos\theta_1\dot{\theta}_1)^2 + (-l_1\sin\theta_1\dot{\theta}_1)^2\right) \\ &+ \frac{1}{2}m_2\left((l_1\cos\theta_1\dot{\theta}_1 + l_2\cos\theta_2\dot{\theta}_2)^2 + (-l_1\sin\theta_1\dot{\theta}_1 - l_2\sin\theta_2\dot{\theta}_2)^2\right) \\ &+ \frac{1}{2}m_3\left((\dot{x}_1 + l_2\cos\theta_2\dot{\theta}_2 + l_3\cos\theta_3\dot{\theta}_3)^2 + (\dot{y}_1 - l_2\sin\theta_2\dot{\theta}_2 - l_3\sin\theta_3\dot{\theta}_3)^2\right) \\ &+ g\left(m_1(y_0 + l_1\cos\theta_1) + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2) + m_3(y_1 + l_2\cos\theta_2 + l_3\cos\theta_3)\right) \end{aligned}$$

$$\begin{aligned} L = &\frac{1}{2}m_1\left((l_1\cos\theta_1\dot{\theta}_1)^2 + (-l_1\sin\theta_1\dot{\theta}_1)^2\right) \\ &+ \frac{1}{2}m_2\left((l_1\cos\theta_1\dot{\theta}_1 + l_2\cos\theta_2\dot{\theta}_2)^2 + (-l_1\sin\theta_1\dot{\theta}_1 - l_2\sin\theta_2\dot{\theta}_2)^2\right) \\ &+ \frac{1}{2}m_3((l_1\cos\theta_1\dot{\theta}_1 + l_2\cos\theta_2\dot{\theta}_2 + l_3\cos\theta_3\dot{\theta}_3)^2 \\ &\qquad + (-l_1\sin\theta_1\dot{\theta}_1 - l_2\sin\theta_2\dot{\theta}_2 - l_3\sin\theta_3\dot{\theta}_3)^2) \\ &+ g(m_1(y_0 + l_1\cos\theta_1) + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2) \\ &\quad + m_3(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2 + l_3\cos\theta_3)) \end{aligned}$$

Expanding the squares:
$(a+b+c)(a+b+c) = a^2+ab+ac+ab+b^2+bc+ac+bc+c^2 = a^2+2ab+b^2+2bc+c^2+2ac$

$$\begin{aligned}
L = &\frac{1}{2}m_1\left(l_1^2\cos^2\theta_1\dot{\theta}_1^2 + l_1^2\sin^2\theta_1\dot{\theta}_1^2\right) \\
&+ \frac{1}{2}m_2(l_1^2\cos^2\theta_1\dot{\theta}_1^2 + 2\cdot l_1\cos\theta_1\dot{\theta}_1 l_2\cos\theta_2\dot{\theta}_2 + l_2^2\cos^2\theta_2\dot{\theta}_2^2 \\
&\qquad + l_1^2\sin^2\theta_1\dot{\theta}_1^2 + 2\cdot l_1\sin\theta_1\dot{\theta}_1 l_2\sin\theta_2\dot{\theta}_2 + l_2^2\sin^2\theta_2\dot{\theta}_2^2) \\
&+ \frac{1}{2}m_3(l_1^2\cos^2\theta_1\dot{\theta}_1^2 + l_2^2\cos^2\theta_2\dot{\theta}_2^2 + l_3^2\cos^2\theta_3\dot{\theta}_3^2 \\
&\qquad + 2\cdot l_1\cos\theta_1\dot{\theta}_1 l_2\cos\theta_2\dot{\theta}_2 + 2\cdot l_1\cos\theta_1\dot{\theta}_1 l_3\cos\theta_3\dot{\theta}_3 \\
&\qquad + 2\cdot l_2\cos\theta_2\dot{\theta}_2 l_3\cos\theta_3\dot{\theta}_3 \\
&\qquad + l_1^2\sin^2\theta_1\dot{\theta}_1^2 + l_2^2\sin^2\theta_2\dot{\theta}_2^2 + l_3^2\sin^2\theta_3\dot{\theta}_3^2 \\
&\qquad + 2\cdot l_1\sin\theta_1\dot{\theta}_1 l_2\sin\theta_2\dot{\theta}_2 + 2\cdot l_1\sin\theta_1\dot{\theta}_1 l_3\sin\theta_3\dot{\theta}_3 \\
&\qquad + 2\cdot l_2\sin\theta_2\dot{\theta}_2 l_3\sin\theta_3\dot{\theta}_3) \\
&+ g(m_1(y_0 + l_1\cos\theta_1) \\
&\quad + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2) \\
&\quad + m_3(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2 + l_3\cos\theta_3))
\end{aligned}$$

$$\begin{aligned}
L = &\frac{1}{2}m_1 l_1^2\dot{\theta}_1^2\left(\cos^2\theta_1 + \sin^2\theta_1\right) \\
&+ \frac{1}{2}m_2(l_1^2\dot{\theta}_1^2(\cos^2\theta_1 + \sin^2\theta_1) + l_2^2\dot{\theta}_2^2(\cos^2\theta_2 + \sin^2\theta_2) \\
&\qquad + 2\cdot l_1 l_2\dot{\theta}_1\dot{\theta}_2(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2)) \\
&+ \frac{1}{2}m_3(l_1^2\dot{\theta}_1^2(\cos^2\theta_1 + \sin^2\theta_1) + l_2^2\dot{\theta}_2^2(\cos^2\theta_2 + \sin^2\theta_2) + l_3^2\dot{\theta}_3^2(\cos^2\theta_3 + \sin^2\theta_3) \\
&\qquad + 2\cdot l_1 l_2\dot{\theta}_1\dot{\theta}_2(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) \\
&\qquad + 2\cdot l_1 l_3\dot{\theta}_1\dot{\theta}_3(\cos\theta_1\cos\theta_3 + \sin\theta_1\sin\theta_3) \\
&\qquad + 2\cdot l_2 l_3\dot{\theta}_2\dot{\theta}_3(\cos\theta_2\cos\theta_3 + \sin\theta_2\sin\theta_3)) \\
&+ g(m_1(y_0 + l_1\cos\theta_1) \\
&\quad + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2) \\
&\quad + m_3(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2 + l_3\cos\theta_3))
\end{aligned}$$

Using $\cos^2 + \sin^2 = 1$ and $\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2 = \cos(\theta_1 - \theta_2)$:

$$\begin{aligned}
L = &\frac{1}{2}m_1 l_1^2\dot{\theta}_1^2 \\
&+ \frac{1}{2}m_2(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2\cdot l_1 l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)) \\
&+ \frac{1}{2}m_3(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + l_3^2\dot{\theta}_3^2 \\
&\qquad + 2\cdot l_1 l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) \\
&\qquad + 2\cdot l_1 l_3\dot{\theta}_1\dot{\theta}_3\cos(\theta_1 - \theta_3) \\
&\qquad + 2\cdot l_2 l_3\dot{\theta}_2\dot{\theta}_3\cos(\theta_2 - \theta_3)) \\
&+ g(m_1(y_0 + l_1\cos\theta_1) \\
&\quad + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2) \\
&\quad + m_3(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2 + l_3\cos\theta_3))
\end{aligned}$$

Euler–Lagrange

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0$$

For θ_1

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) - \frac{\partial L}{\partial \theta_1} = 0$$

$$\begin{aligned}
\frac{\partial L}{\partial \theta_1} = \frac{\partial}{\partial \theta_1}\Bigg[&\frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 \\
&+ \frac{1}{2}m_2\big(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2\cdot l_1 l_2 \dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)\big) \\
&+ \frac{1}{2}m_3\big(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + l_3^2\dot{\theta}_3^2 \\
&\qquad + 2\cdot l_1 l_2 \dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2) \\
&\qquad + 2\cdot l_1 l_3 \dot{\theta}_1\dot{\theta}_3\cos(\theta_1-\theta_3) \\
&\qquad + 2\cdot l_2 l_3 \dot{\theta}_2\dot{\theta}_3\cos(\theta_2-\theta_3)\big) \\
&+ g\big(m_1(y_0 + l_1\cos\theta_1) \\
&\quad + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2) \\
&\quad + m_3(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2 + l_3\cos\theta_3)\big)\Bigg]
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial \theta_1} = &\frac{1}{2}m_2\big(-2\cdot l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2)\big) \\
&+ \frac{1}{2}m_3\big(-2\cdot l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
&\qquad - 2\cdot l_1 l_3\dot{\theta}_1\dot{\theta}_3\sin(\theta_1-\theta_3)\big) \\
&+ g\big(-m_1(l_1\sin\theta_1) - m_2(l_1\sin\theta_1) - m_3(l_1\sin\theta_1)\big)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial \theta_1} = &-m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
&- m_3\big(l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
&\qquad + l_1 l_3\dot{\theta}_1\dot{\theta}_3\sin(\theta_1-\theta_3)\big) \\
&- g l_1(m_1+m_2+m_3)\sin\theta_1
\end{aligned}$$

$$
\begin{aligned}
\frac{\partial L}{\partial \dot\theta_1} = \frac{\partial}{\partial \dot\theta_1}\Bigg[\frac{1}{2}m_1 l_1^2\dot\theta_1^2
&+ \frac{1}{2}m_2\big(l_1^2\dot\theta_1^2 + l_2^2\dot\theta_2^2 + 2\cdot l_1 l_2\dot\theta_1\dot\theta_2\cos(\theta_1-\theta_2)\big)\\
&+ \frac{1}{2}m_3\big(l_1^2\dot\theta_1^2 + l_2^2\dot\theta_2^2 + l_3^2\dot\theta_3^2\\
&\qquad + 2\cdot l_1 l_2\dot\theta_1\dot\theta_2\cos(\theta_1-\theta_2)\\
&\qquad + 2\cdot l_1 l_3\dot\theta_1\dot\theta_3\cos(\theta_1-\theta_3)\\
&\qquad + 2\cdot l_2 l_3\dot\theta_2\dot\theta_3\cos(\theta_2-\theta_3)\big)\\
&+ g\big(m_1(y_0 + l_1\cos\theta_1)\\
&\quad + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2)\\
&\quad + m_3(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2 + l_3\cos\theta_3)\big)\Bigg]
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \dot\theta_1} &= m_1 l_1^2\dot\theta_1\\
&+ \frac{1}{2}m_2\big(2\cdot l_1^2\dot\theta_1 + 2\cdot l_1 l_2\dot\theta_2\cos(\theta_1-\theta_2)\big)\\
&+ \frac{1}{2}m_3\big(2\cdot l_1^2\dot\theta_1 + 2\cdot l_1 l_2\dot\theta_2\cos(\theta_1-\theta_2) + 2\cdot l_1 l_3\dot\theta_3\cos(\theta_1-\theta_3)\big)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \dot\theta_1} &= m_1 l_1^2\dot\theta_1\\
&+ m_2\big(l_1^2\dot\theta_1 + l_1 l_2\dot\theta_2\cos(\theta_1-\theta_2)\big)\\
&+ m_3\big(l_1^2\dot\theta_1 + l_1 l_2\dot\theta_2\cos(\theta_1-\theta_2) + l_1 l_3\dot\theta_3\cos(\theta_1-\theta_3)\big)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\theta_1}\right) = \frac{d}{dt}\Big[m_1 l_1^2\dot\theta_1
&+ m_2\big(l_1^2\dot\theta_1 + l_1 l_2\dot\theta_2\cos(\theta_1-\theta_2)\big)\\
&+ m_3\big(l_1^2\dot\theta_1 + l_1 l_2\dot\theta_2\cos(\theta_1-\theta_2) + l_1 l_3\dot\theta_3\cos(\theta_1-\theta_3)\big)\Big]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\theta_1}\right) &= m_1 l_1^2\ddot\theta_1\\
&+ m_2\big(l_1^2\ddot\theta_1 + l_1 l_2(\ddot\theta_2\cos(\theta_1-\theta_2) - \dot\theta_2\sin(\theta_1-\theta_2)(\dot\theta_1-\dot\theta_2))\big)\\
&+ m_3\big(l_1^2\ddot\theta_1\\
&\qquad + l_1 l_2(\ddot\theta_2\cos(\theta_1-\theta_2) - \dot\theta_2\sin(\theta_1-\theta_2)(\dot\theta_1-\dot\theta_2))\\
&\qquad + l_1 l_3(\ddot\theta_3\cos(\theta_1-\theta_3) - \dot\theta_3\sin(\theta_1-\theta_3)(\dot\theta_1-\dot\theta_3))\big)
\end{aligned}
$$

$$
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\theta_1}\right) - \frac{\partial L}{\partial \theta_1} = 0
$$

$$
\begin{aligned}
0 = {} & m_1 l_1^2 \ddot{\theta}_1 \\
& + m_2\Big(l_1^2\ddot{\theta}_1 + l_1 l_2\big[\ddot{\theta}_2\cos(\theta_1-\theta_2) - \dot{\theta}_2\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\big]\Big) \\
& + m_3\Big(l_1^2\ddot{\theta}_1 \\
& \qquad + l_1 l_2\big[\ddot{\theta}_2\cos(\theta_1-\theta_2) - \dot{\theta}_2\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\big] \\
& \qquad + l_1 l_3\big[\ddot{\theta}_3\cos(\theta_1-\theta_3) - \dot{\theta}_3\sin(\theta_1-\theta_3)(\dot{\theta}_1-\dot{\theta}_3)\big]\Big) \\
& - \Bigg(- m_2 l_1 l_2 \dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& \qquad - m_3\Big[l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& \qquad\qquad + l_1 l_3\dot{\theta}_1\dot{\theta}_3\sin(\theta_1-\theta_3)\Big] \\
& \qquad - g l_1(m_1+m_2+m_3)\sin\theta_1\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
0 = {} & m_1 l_1^2\ddot{\theta}_1 \\
& + m_2 l_1^2\ddot{\theta}_1 + m_2 l_1 l_2\ddot{\theta}_2\cos(\theta_1-\theta_2) - m_2 l_1 l_2\dot{\theta}_2\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2) \\
& + m_3 l_1^2\ddot{\theta}_1 \\
& + m_3 l_1 l_2\ddot{\theta}_2\cos(\theta_1-\theta_2) - m_3 l_1 l_2\dot{\theta}_2\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2) \\
& + m_3 l_1 l_3\ddot{\theta}_3\cos(\theta_1-\theta_3) - m_3 l_1 l_3\dot{\theta}_3\sin(\theta_1-\theta_3)(\dot{\theta}_1-\dot{\theta}_3) \\
& + m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_3 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_3 l_1 l_3\dot{\theta}_1\dot{\theta}_3\sin(\theta_1-\theta_3) \\
& + g l_1(m_1+m_2+m_3)\sin\theta_1
\end{aligned}
$$

$$
\begin{aligned}
0 = {} & m_1 l_1^2\ddot{\theta}_1 + m_2 l_1^2\ddot{\theta}_1 + m_3 l_1^2\ddot{\theta}_1 \\
& + m_2 l_1 l_2\ddot{\theta}_2\cos(\theta_1-\theta_2) \\
& - m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_2 l_1 l_2\dot{\theta}_2\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_3 l_1 l_2\ddot{\theta}_2\cos(\theta_1-\theta_2) \\
& - m_3 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_3 l_1 l_2\dot{\theta}_2\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_3 l_1 l_3\ddot{\theta}_3\cos(\theta_1-\theta_3) \\
& - m_3 l_1 l_3\dot{\theta}_1\dot{\theta}_3\sin(\theta_1-\theta_3) \\
& + m_3 l_1 l_3\dot{\theta}_3\dot{\theta}_3\sin(\theta_1-\theta_3) \\
& + m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_3 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) \\
& + m_3 l_1 l_3\dot{\theta}_1\dot{\theta}_3\sin(\theta_1-\theta_3) \\
& + g l_1(m_1+m_2+m_3)\sin\theta_1
\end{aligned}
$$

$$
\begin{aligned}
0 = {} & m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_3 l_1^2 \ddot{\theta}_1 \\
& + m_2 \Big[l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& \qquad + l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \Big] \\
& + m_3 \Big[l_1 l_3 \ddot{\theta}_3 \cos(\theta_1 - \theta_3) \\
& \qquad + l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& \qquad + l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\
& \qquad + l_1 l_3 \dot{\theta}_3^2 \sin(\theta_1 - \theta_3) \Big] \\
& + g l_1 (m_1 + m_2 + m_3) \sin\theta_1
\end{aligned}
$$

Solution for θ_1

$$
\begin{aligned}
(m_1 + m_2 + m_3) l_1^2 \ddot{\theta}_1 + (m_2 + m_3) l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
+ m_3 l_1 l_3 \ddot{\theta}_3 \cos(\theta_1 - \theta_3) = (-m_2 - m_3) l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\
- m_3 l_1 l_3 \dot{\theta}_3^2 \sin(\theta_1 - \theta_3) - g l_1 (m_1 + m_2 + m_3) \sin\theta_1
\end{aligned}
$$

For θ_2

$$
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) - \frac{\partial L}{\partial \theta_2} = 0
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \theta_2} = \frac{\partial}{\partial \theta_2} \Bigg[& \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 \\
& + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\
& + \frac{1}{2} m_3 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + l_3^2 \dot{\theta}_3^2 \\
& \qquad + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& \qquad + 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \cos(\theta_1 - \theta_3) \\
& \qquad + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \cos(\theta_2 - \theta_3)) \\
& + g \big(m_1 (y_0 + l_1 \cos\theta_1) \\
& \qquad + m_2 (y_0 + l_1 \cos\theta_1 + l_2 \cos\theta_2) \\
& \qquad + m_3 (y_0 + l_1 \cos\theta_1 + l_2 \cos\theta_2 + l_3 \cos\theta_3) \big) \Bigg]
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \theta_2} = & \frac{1}{2} m_2 \Big(2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) \cdot (-1) \Big) \\
& + \frac{1}{2} m_3 \Big(2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) \cdot (-1) \\
& \qquad + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 (-\sin(\theta_2 - \theta_3)) \cdot (-1) \Big) \\
& + g \Big(m_2 l_2 (-\sin\theta_2) + m_3 l_2 (-\sin\theta_2) \Big)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \theta_2} = \quad & m_2 l_1 l_2 \dot\theta_1 \dot\theta_2 \sin(\theta_1 - \theta_2) \\
& + m_3\Big(l_1 l_2 \dot\theta_1 \dot\theta_2 \sin(\theta_1 - \theta_2) \\
& \qquad + l_2 l_3 \dot\theta_2 \dot\theta_3 \sin(\theta_2 - \theta_3)\Big) \\
& \quad + g\Big(m_2 l_2(-\sin\theta_2) + m_3 l_2(-\sin\theta_2)\Big)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \dot\theta_2} = \frac{\partial}{\partial \dot\theta_2}\Bigg[&\frac{1}{2} m_1 l_1^2 \dot\theta_1^2 \\
& + \frac{1}{2} m_2 (l_1^2 \dot\theta_1^2 + l_2^2 \dot\theta_2^2 + 2 \cdot l_1 l_2 \dot\theta_1 \dot\theta_2 \cos(\theta_1 - \theta_2)) \\
& + \frac{1}{2} m_3 (l_1^2 \dot\theta_1^2 + l_2^2 \dot\theta_2^2 + l_3^2 \dot\theta_3^2 \\
& \qquad + 2 \cdot l_1 l_2 \dot\theta_1 \dot\theta_2 \cos(\theta_1 - \theta_2) \\
& \qquad + 2 \cdot l_1 l_3 \dot\theta_1 \dot\theta_3 \cos(\theta_1 - \theta_3) \\
& \qquad + 2 \cdot l_2 l_3 \dot\theta_2 \dot\theta_3 \cos(\theta_2 - \theta_3)) \\
& + g(m_1(y_0 + l_1 \cos\theta_1) \\
& \quad + m_2(y_0 + l_1 \cos\theta_1 + l_2 \cos\theta_2) \\
& \quad + m_3(y_0 + l_1 \cos\theta_1 + l_2 \cos\theta_2 + l_3 \cos\theta_3))\Bigg]
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \dot\theta_2} = \quad & \frac{1}{2} m_2 (2 \cdot l_2^2 \dot\theta_2 + 2 \cdot l_1 l_2 \dot\theta_1 \cos(\theta_1 - \theta_2)) \\
& + \frac{1}{2} m_3 \Big(2 \cdot l_2^2 \dot\theta_2 \\
& \qquad + 2 \cdot l_1 l_2 \dot\theta_1 \cos(\theta_1 - \theta_2) \\
& \qquad + 2 \cdot l_2 l_3 \dot\theta_3 \cos(\theta_2 - \theta_3)\Big)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \dot\theta_2} = \quad & m_2 (l_2^2 \dot\theta_2 + l_1 l_2 \dot\theta_1 \cos(\theta_1 - \theta_2)) \\
& + m_3 \Big(l_2^2 \dot\theta_2 + l_1 l_2 \dot\theta_1 \cos(\theta_1 - \theta_2) + l_2 l_3 \dot\theta_3 \cos(\theta_2 - \theta_3)\Big)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\theta_1}\right) = \frac{d}{dt}\Big[& m_2 (l_2^2 \dot\theta_2 + l_1 l_2 \dot\theta_1 \cos(\theta_1 - \theta_2)) \\
& + m_3 \Big(l_2^2 \dot\theta_2 + l_1 l_2 \dot\theta_1 \cos(\theta_1 - \theta_2) + l_2 l_3 \dot\theta_3 \cos(\theta_2 - \theta_3)\Big)\Big]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\theta_1}\right) = & m_2 \Big(l_2^2 \ddot\theta_2 + l_1 l_2 (\ddot\theta_1 \cos(\theta_1 - \theta_2) - \dot\theta_1 \sin(\theta_1 - \theta_2)(\dot\theta_1 - \dot\theta_2))\Big) \\
& + m_3 \Big(l_2^2 \ddot\theta_2 \\
& \quad + l_1 l_2 (\ddot\theta_1 \cos(\theta_1 - \theta_2) - \dot\theta_1 \sin(\theta_1 - \theta_2)(\dot\theta_1 - \dot\theta_2)) \\
& \quad + l_2 l_3 (\ddot\theta_3 \cos(\theta_2 - \theta_3) - \dot\theta_3 \sin(\theta_2 - \theta_3)(\dot\theta_2 - \dot\theta_3))\Big)
\end{aligned}
$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) - \frac{\partial L}{\partial \theta_1} = 0$$

$$\begin{aligned}
0 = \quad & m_2\Big(l_2^2\ddot{\theta}_2 + l_1 l_2\big(\ddot{\theta}_1\cos(\theta_1-\theta_2) - \dot{\theta}_1\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\big)\Big)\\
& + m_3\Big(l_2^2\ddot{\theta}_2\\
& \qquad + l_1 l_2\big(\ddot{\theta}_1\cos(\theta_1-\theta_2) - \dot{\theta}_1\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\big)\\
& \qquad + l_2 l_3\big(\ddot{\theta}_3\cos(\theta_2-\theta_3) - \dot{\theta}_3\sin(\theta_2-\theta_3)(\dot{\theta}_2-\dot{\theta}_3)\big)\Big)\\
& - \Bigg(m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) + m_3\Big(l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2)\\
& \qquad + l_2 l_3\dot{\theta}_2\dot{\theta}_3\sin(\theta_2-\theta_3)\Big) + g\Big(m_2 l_2(-\sin\theta_2) + m_3 l_2(-\sin\theta_2)\Big)\Bigg)
\end{aligned}$$

$$\begin{aligned}
0 = \quad & m_2 l_2^2\ddot{\theta}_2 + m_2 l_1 l_2\big(\ddot{\theta}_1\cos(\theta_1-\theta_2) - \dot{\theta}_1\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\big)\\
& + m_3 l_2^2\ddot{\theta}_2 + m_3 l_1 l_2\big(\ddot{\theta}_1\cos(\theta_1-\theta_2) - \dot{\theta}_1\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\big)\\
& \qquad + m_3 l_2 l_3\big(\ddot{\theta}_3\cos(\theta_2-\theta_3) - \dot{\theta}_3\sin(\theta_2-\theta_3)(\dot{\theta}_2-\dot{\theta}_3)\big)\\
& - m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) - m_3\Big(l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2)\\
& \qquad - l_2 l_3\dot{\theta}_2\dot{\theta}_3\sin(\theta_2-\theta_3)\Big) - g\Big(m_2 l_2(-\sin\theta_2) + m_3 l_2(-\sin\theta_2)\Big)
\end{aligned}$$

$$\begin{aligned}
0 = \quad & m_2 l_2^2\ddot{\theta}_2\\
& + m_2 l_1 l_2\ddot{\theta}_1\cos(\theta_1-\theta_2)\\
& - m_2 l_1 l_2\dot{\theta}_1\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\\
& + m_3 l_2^2\ddot{\theta}_2\\
& + m_3 l_1 l_2\ddot{\theta}_1\cos(\theta_1-\theta_2)\\
& - m_3 l_1 l_2\dot{\theta}_1\sin(\theta_1-\theta_2)(\dot{\theta}_1-\dot{\theta}_2)\\
& + m_3 l_2 l_3\ddot{\theta}_3\cos(\theta_2-\theta_3)\\
& - m_3 l_2 l_3\dot{\theta}_3\sin(\theta_2-\theta_3)(\dot{\theta}_2-\dot{\theta}_3)\\
& - m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2)\\
& - m_3 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2)\\
& + m_3 l_2 l_3\dot{\theta}_2\dot{\theta}_3\sin(\theta_2-\theta_3)\\
& + g(m_2+m_3)l_2(\sin\theta_2)
\end{aligned}$$

$$
\begin{aligned}
0 = \quad & m_2 l_2^2 \ddot{\theta}_2 \\
& + m_2 l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) \\
& - m_2 l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) \\
& + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& - m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_2^2 \ddot{\theta}_2 \\
& + m_3 l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) \\
& - m_3 l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& - m_3 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_2 l_3 \ddot{\theta}_3 \cos(\theta_2 - \theta_3) \\
& + m_3 l_2 l_3 \dot{\theta}_3^2 \sin(\theta_2 - \theta_3) \\
& + m_3 l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \sin(\theta_2 - \theta_3) \\
& - m_3 l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \sin(\theta_2 - \theta_3) \\
& + g(m_2 + m_3) l_2 (\sin\theta_2)
\end{aligned}
$$

$$
\begin{aligned}
0 = \quad & l_2^2 \ddot{\theta}_2 (m_2 + m_3) \\
& + l_1 l_2 \ddot{\theta}_1 (m_2 + m_3) \cos(\theta_1 - \theta_2) \\
& + m_3 l_2 l_3 \ddot{\theta}_3 \cos(\theta_2 - \theta_3) \\
& - l_1 l_2 (m_2 + m_3) \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_2 l_3 \dot{\theta}_3^2 \sin(\theta_2 - \theta_3) \\
& + g(m_2 + m_3) l_2 (\sin\theta_2)
\end{aligned}
$$

$$
\begin{aligned}
& l_2^2 \ddot{\theta}_2 (m_2 + m_3) + l_1 l_2 \ddot{\theta}_1 (m_2 + m_3) \cos(\theta_1 - \theta_2) \\
+ & m_3 l_2 l_3 \ddot{\theta}_3 \cos(\theta_2 - \theta_3) = l_1 l_2 (m_2 + m_3) \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) \\
- & m_3 l_2 l_3 \dot{\theta}_3^2 \sin(\theta_2 - \theta_3) - g(m_2 + m_3) l_2 (\sin\theta_2)
\end{aligned}
$$